✎ Evolution of Cooperation in Stochastic Games

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Abstract

Social dillemas arise when individual incentives conflict with collective wellfare, leading to depletion of shared ressources. Classical analyses typicaly rely on repeated games with fixed payoffs, neglecting the fact that strategic behaviour can itself modify the environment and thereby reshape future incentives. The main goal of this paper is reproduce and analyze the figures of the paper. This reproduction successfully captures several of the main figures reported in the original paper, including some extended figures that will be explained in this paper. At the same time, some quantitative discrepancies arised in the replication of some figures, highlighting sensitivities to modelling choices and implemantations details. Overall, this work confirms the robustness of the core mechanism promoting cooperations while illustrating while illustrating the inherited challenges due to reproducing complex stochastic evolutionnary models.

1 Introduction

In evolutionary game theory, the simple question about understanding how cooperation emerges and persists in social dilemmas is crucial. Classical models analyse repeated interactions in environmments where the payoffs structure remains fixed across rounds. In this case, it leads to a limited view of strategic behaviour. In fact, cooperation can be sustained only through direct reciprocity, and only under restrictive conditions on discounting, noise, or strategic complexity. Such models omit a central characteristic of real-world dilemmas, namely that strategic behaviour feeds back into and alters the environment itself. Cooperative behaviour may improve shared resources, where defection may degrade them, thereby reshaping the incentives that govern subsequent interactions. The paper reproduced in this report, *"Evolution of Cooperation in Stochastic Games"* by Hilbe, Šimsa, Chatterjee and Nowak (2018) introduces an approach that explicitly couples strategic behaviour with environmental dynamics. The authors model social dilemmas as stochastic games in which the state of the environment determines the payoffs of the current round, while the actiones of the playes determine transitions between states. It permits to generalize repeated games by allowing cooperation or defection to influence the value of future interactions. Their contribution was to prove that such feeback can dramatically enhance coopreation, even in cases where all individual repeated gamous would favour defection. The paper demonstrates, through evolutionary simulations, that cooperation becomes advantageous when mutual cooperation leads to more profitable future states and that mutual defection leads to deterioration. One of the methods used was *win-stay lose-shift* (WSLS), which is not generally stable in classical repeated games, and it becomes that dominant evolutionary outcome once cooperation restores the high-value state and defection drives the system toward deterioation. Numerical analyses confim that WSLS leads to full cooperation in the two-player case under these feedback dynamics. In the rest of the paper, the authorsexamine deterministic and probabilistic transitions, multi-state public-goods environments, and different forms of envionmental degradation. Across all scenarios, coupling behaviour to environmental change consistently promotes cooperation, even when each constituent repeated game alone favours defection. Reproducing all these results is important an necessary. It validates the robustness of the authors' claims and the reproducibility of their computational approach. It also provides deeper insight into how environmental feedback mechanisms can mitigate the tragedy of the commons by making defection immediately detrimental to the shared environment. This report presents the reproduction of the main figures and results of the paper *"Evolution of Cooperation in Stochastic Games"*, along with an analysis of the underlying dynamics and the conditions under which cooperation is most strongly promoted.

2 Related Work

Research on cooperation in dynamic environments has expanded significantly in recent years, particularly through models that couple strategic behaviour with environmental change. Below, we discuss three papers closely related to the one reproduced in this report, both in terms of objectives and methodology.

Dynamics of cooperation in concurrent games *[1].* This paper develops a theoretical and experimental framework to analyse cooperation when individuals engage in several repeated games *simultaneously*. The authors introduce the notion of *concurrent games*, where behaviour in one interaction can influence behaviour in another through strategic spillovers or cognitive constraints such as imperfect recall. Using evolutionary learning dynamics and a behavioural experiment with human participants, the study shows that cross-game linkage can substantially alter cooperation levels. In particular, cooperation tends to decrease when players participate in multiple games in parallel, especially when partners differ across games, highlighting how strategic motives and behavioural spillovers reshape reciprocity.

Difference with the reproduced paper. Unlike the paper reproduced in this report, which focuses on how environmental feedback within a single stochastic game promotes cooperation, Rossetti et al. investigate a fundamentally different mechanism: the interaction between multiple concurrent games. Their work does not involve state-dependent payoffs or environmental transitions; instead, it examines how multi-game structure, linkage, and cognitive constraints affect the evolution of cooperation. Although both papers rely on evolutionary game-theoretic tools, they address distinct sources of complexity in cooperative behaviour.

Evolutionary dynamics in stochastic nonlinear public goods games *[2].* This paper develops a probabilistic framework for public goods games in which group interactions exhibit *stochastic*

nonlinearity. Instead of assuming fixed synergy or discounting effects, each group independently displays either synergistic or diminishing-returns benefits with a given probability. Using analytical derivations and Monte Carlo simulations in both well-mixed and structured populations, the authors show that increasing the degree of nonlinearity can amplify the advantage of synergy, thereby promoting cooperation, while also revealing cases where network reciprocity becomes ineffective. Their results highlight how uncertainty in group-level payoff generation fundamentally alters cooperative dynamics.

Difference with the reproduced paper. Unlike the paper reproduced in this report, which studies how *state transitions driven by players' actions* shape cooperation in stochastic games, Zhu et al. investigate a different source of stochasticity: *random nonlinear payoff structures* within public goods groups. Their model does not involve strategic feedback loops between behaviour and environmental states, nor memory-one reciprocity. Instead, it focuses on how probabilistic synergy and discounting influence evolutionary outcomes in multi-player interactions. While both works analyse cooperation under environmental uncertainty, they address distinct mechanisms and operate in different game-theoretic settings.

The effect of environmental information on evolution of cooperation in stochastic games: *[3].* This paper extends the stochastic- game approach by examining how the availability of information about the current environmental state affects the evolution of cooperation. The authors compare two settings: a *full-information* scenario, where players can condition their strategies on the observed state, and a *no-information* scenario, where players must act identically across states. They introduce the notion of a *value of information*, quantifying how cooperation levels change when state information becomes available. Their systematic analysis reveals that information is often beneficial but can, in some cases, reduce cooperation—leading to a "benefit of ignorance."

Difference with the reproduced paper. Unlike the paper reproduced in this report, which investigates how environmental feedback itself promotes cooperation, Kleshnina et al. focus on the role of *information* rather than feedback. Their contribution lies in analysing how uncertainty about the state modifies evolutionary outcomes, whereas the reproduced paper studies how state transitions driven by behaviour create incentives for cooperation. Thus, the two works share the same modelling framework but address fundamentally different questions.

3 Practical aspects of the reproduction

The original paper provided MATLAB code for Figures 2, 3, and 4. However, since no one in our group know how to code in MATLAB, we chose to implement the entire project using Python. We used the provided MATLAB code only as a reference to understand the logic, but we wrote our own code from scratch.

For the **Extended Data figures**, there was no code provided. To reproduce these, we had to rely entirely on the mathematical formulas found in the Supplementary Information and build the simulation logic ourselves.

Initially, we implemented the simulation using multiple separate Python files. However, we realized that this workflow made it difficult to view and analyze the resulting plots immediately. To address this, we integrated our code into **Jupyter Notebooks**. This transition allowed us to visualize the figures directly within the document alongside the code.

Within the notebook, we organized the logic into three clear parts:

(1) **Game Logic:** A module that calculates the payoff scores and handles the rules for environment switching.
(2) **Evolutionary Process:** A separate function to calculate the "fixation probability" (who wins in the population). Separating this allowed us to easily switch between different simulation methods (Global vs. Local mutation) required for different figures.
(3) **Parallel Processing:** Since the simulation requires averaging thousands of runs, the process was computationally expensive. We used Python's `multiprocessing` library to run many simulations at the same time, which significantly reduced the runtime.

We faced several challenges during the reproduction process:

- **Ambiguity regarding the Mutation Rate:** The paper describes the model as operating under "rare mutations" but does not explicitly state the specific numerical value used in the simulations. We experimented with different values, specifically testing 0.001 and 0.0001. Ultimately, we adopted a rate of 0.001, as we found this value produced results that best matched the figures in the original paper.

4 Methods

4.1 Usage of AI

In the process of producing the results, we used AI Models, specifically ChatGPT-4. An honest assessment of their contribution follows:

- **Theoretical Clarification:** Given the complexity of the stochastic game framework, we used AI tools to accelerate our understanding of key theoretical concepts, specifically the construction of state-transition matrices for memory-one strategies.
- **Code Debugging:** AI tools was used in identifying the source of numerical instability (overflow errors) in the fixation probability calculations.
- **Performance Optimization:** Our initial Python implementation suffered from slow execution speeds. We asked AI tools for optimization strategies, and they recommended using the `numba` library. Following this recommendation, we studied the underlying mechanics of `numba` to understand how it optimizes Python bytecode and successfully integrated it to significantly accelerate our core simulation loops.
- **Data Visualization:** We asked AI tool to query specific functions within the `matplotlib` library for generating the required figures. AI assisted in adjusting plot parameters (such as heatmaps and trajectory lines) to ensure the visual style matched the original paper, but the data fed into these functions was generated and verified by our own code.

Assessment: AI tools functioned solely as **technical consultants**. Most of the time it gave us wrong answer, but they did

provided suggestions and optimization strategies, the **core scientific contributions** were exclusively human.(like The logical implementation of the game dynamics,decision-making regarding parameter values,the understanding of optimization mechanics). The final verification of all results were done by us.

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5 Results

5.1 Evolutionary Trajectories in Stochastic Games(Fig2a)

We simulated the evolutionary dynamics of cooperation under environmental feedback. We aimed to replicate the core finding of the original paper: that linking game payoffs to past actions can stabilize cooperation in the Prisoner's Dilemma.

We ran agent-based simulations for $N = 100$ individuals over 5,000 time steps, averaging the results across 100 independent runs.The parameters were set to match Figure 2a of the original paper: benefit $b_1 = 2.0$ (State 1), $b_2 = 1.2$ (State 2), and cost $c = 1.0$.

5.1.1 Comparison of Stochastic and Static Environments. Figure **??** presents the results generated by our simulation code.

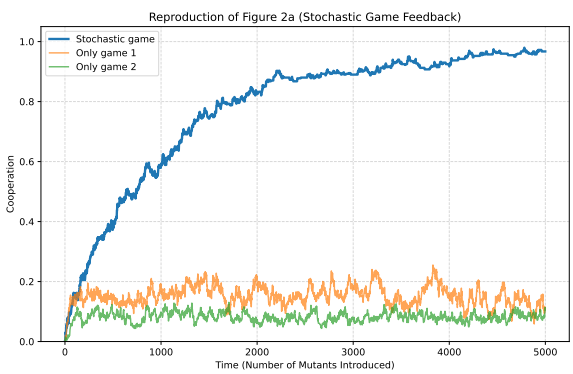

Figure 1: Reproduction of Figure 2a.

The trajectories clearly demonstrate the effectiveness of environmental feedback:

- **Stochastic Game (Blue Line):** Starting from a population of defectors, cooperation emerges quickly. As defined in our function get_payoff_and_coop, players only access the beneficial state (Game 1) through mutual cooperation.This mechanism imposes a double penalty on defectors: they lose immediate payoff and degrade their future environment. Consequently, the cooperation rate rises to approximately 95% within 3,000 steps.
- **Static Controls (Orange/Green Lines):** When we ran the simulation with the environment fixed ("Only Game 1" or "Only Game 2"), cooperation failed to evolve, they fluctuate around 10-20%. This means that high benefits alone ($b = 2.0$) are insufficient to sustain cooperation without the coupling mechanism.

These results are identical to the original Figure 2a.

5.2 Multiplayer Public Goods Game (Figure 2b)

We extended our analysis to a four-player Public Goods Game (Group Size = 4). And we found out that simulating group interactions is computationally more intensive than pairwise interactions; so, we used Numba to speed up.

Based on the group parameters from the paper, we set the multiplication factors to $r_1 = 1.6$ (High Payoff) and $r_2 = 1.2$ (Low Payoff). To observe clear evolutionary trends within a reasonable timeframe (10,000 generations), we used a stronger selection intensity of $\beta = 100.0$(in the pairwise case we used $\beta = 1.0$)

Figure2b presents the evolutionary trajectories averaged over 100 independent simulation runs.

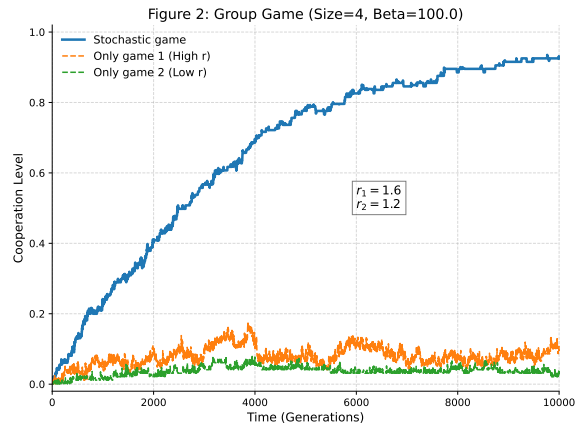

Figure 2: Reproduction of Figure 2b (Group Interaction).

5.2.1 Observations and Analysis.

- **Robust Cooperation in Groups:** As shown in the figure (blue line), cooperation in the stochastic group game rises and reached (> 90%) after approximately 7500 generations. This means that the feedback mechanism remains effective even when the social dilemma involves multiple actors.
- **Validation of Transition Logic:** The success of this trajectory validates the specific transition rule implemented in our code. We set the transition vector (q_{vec}) so that the environment only improves if everyone cooperates. If even one person defects, the state stays the same. This rule forces the group to work together and prevents people from cheating.
- **Effect of Selection Intensity (β):** A key observation from our reproduction is the necessity of strong selection in group games. While $\beta = 1.0$ was sufficient for pairwise games (Fig 2a), our simulations confirmed that a higher intensity ($\beta = 100.0$) is critical for the emergence of cooperation in the 4-player setting. This is because in a bigger group, one person's actions matter less to the whole resource, so we need a stricter environment to stop people from cheating.
- **Controls:** Just like in the two-player games, the static tests (orange and green lines) show that cooperation fails. This proves that if the environment doesn't change and react to

how people play, everyone ends up being selfish, and the shared resource is destroyed.

5.3 Probabilistic transitions maximize cooperation in three different stochastic games.

5.3.1 Probabilistic Return to a Beneficial State. In this scenario, players interact in two states: Game 1 (high benefit) and Game 2 (low benefit). Mutual cooperation in the inferior Game 2 leads back to the superior Game 1 with probability q.

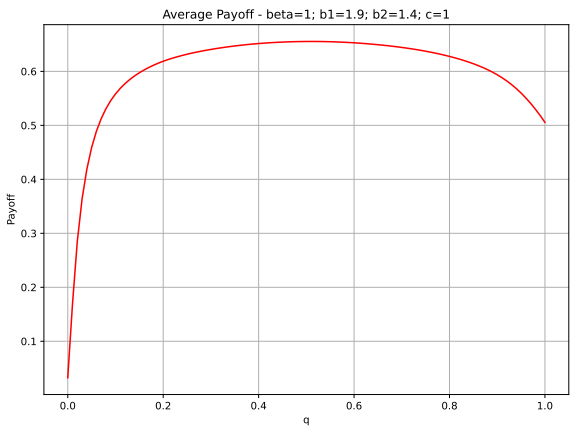

Figure 3: Reproduction figure 3a

We can deduce that when q is very low, it is nearly impossible to return to the beneficial state once lost, trapping the population in the low-payoff game. Vice versa, when q approaches 1, the return to the beneficial state is guaranteed, reducing the penalty for defection and allowing exploitation. Intermediate values of q provide enough incentive to cooperate to regain the good state without making the recovery too easy. Infact the reproduced curve shows that the average payoff is maximized at an intermediate value.

5.3.2 Endogenous Game Termination. This figure analyzes a scenario where the game length is determined by player behavior. Players start in a profitable state, but if anyone defects, there is a probability q of moving to a state with no profitable interactions (effectively ending the game).

A small probability of the game ending acts as a powerful deterrent against defection, enforcing cooperation. However, as q becomes too high, the game becomes unstable; even small error rates ($epsilon$) cause the game to terminate prematurely, reducing the total accumulated payoff.

5.3.3 Evolutionary Dynamics with Timeouts. This panel presents the phase portrait for a game involving a "timeout" state. Defection leads to a temporary state where no interaction occurs, and players return to the game with probability q. The axes represent the probability of cooperating in the regular state (x) versus after a timeout (y).

We can observe that the vector field shows the evolutionary trajectories of the population. The flow indicates that for specific

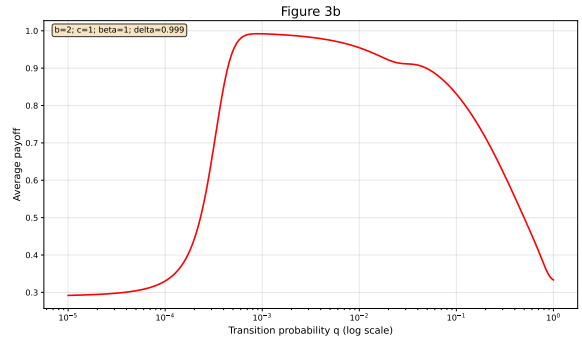

Figure 4: Reproduction figure 3b

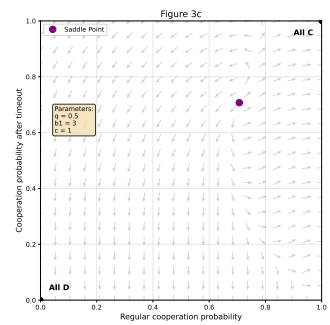

Figure 5: Reproduction figure 3c

parameters, the population evolves away from defection and converges toward the "All C" corner (1, 1).
So the reproduction confirms that fully cooperative strategies (All C) can become stable endpoints of evolution in this stochastic setting. This contrasts with standard repeated Prisoner's Dilemma games where unconditional cooperation is generally not stable against invasion by defectors.

5.4 Immediate environmental feedback enhances cooperation

This panel plots the multiplication factor r across five environmental states for four different feedback scenarios6.

- Immediate (Blue): The resource value drops sharply after the first deterioration step, creating strong pressure to maintain the optimal state.
- Gradual (Green): The resource degrades linearly as the state worsens.
- Delayed (Orange): The resource remains valuable for most states and only collapses when the environment is nearly ruined.
- None (Red): The resource quality remains constant (r = 1.6) regardless of the environmental state, simulating a traditional repeated game without environmental feedback.

5.4.1 Evolutionary analysis. Looking at figure 7, immediate scenario achieves the highest cooperation rate (0.90), drastically outperforming the None scenario.

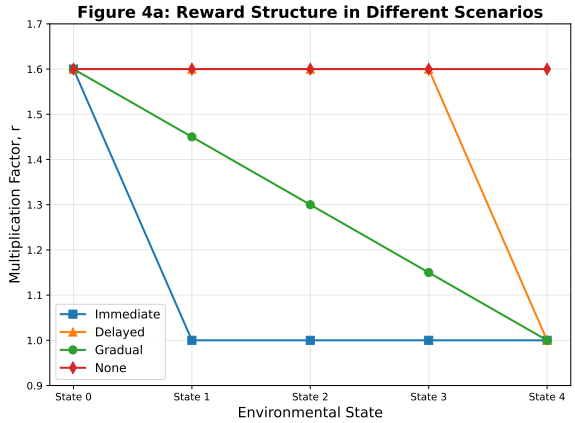

Figure 6: Reproduction figure 4a

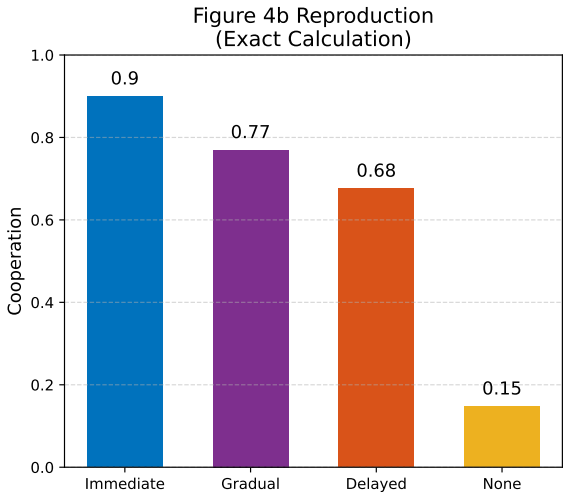

Figure 7: Reproduction figure 4b

So when the environment reacts immediately to exploitation (defection), players are incentivized to cooperate to avoid a steep reduction in future payoffs. Even a Delayed response (0.68) is significantly better than no feedback at all, illustrating that any dependency between resource quality and player behavior helps mitigate the social dilemma.

5.5 Extended Figure 1:

In general, the figure from a to d were successfully replicated.

In both pure and stochastic strategy approaches, we can observe that the green line (representing the stochastic game) maintains a high level of cooperation even at lower benefit values where the red line (representing only the game 2) callapses. This confirms the results of the paper that the prospect of playing a more valuable game significantly enhances the propensity for cooperation, especially when the benefit is in an intermediate range. The important thing to notice in this reproduction, it is that the mechanism are highly sensitive to noise. As the error rate increases, the probability

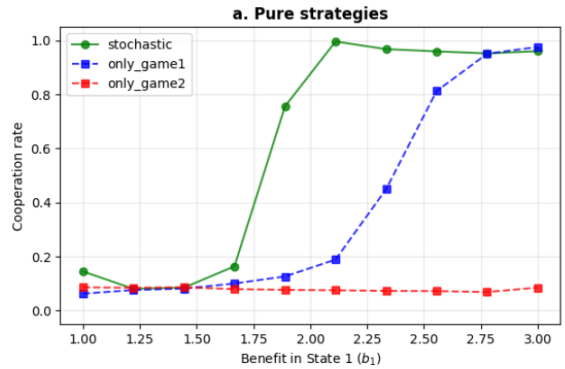

Figure 8: Reproduction of Pure Strategies with benefit in State 1

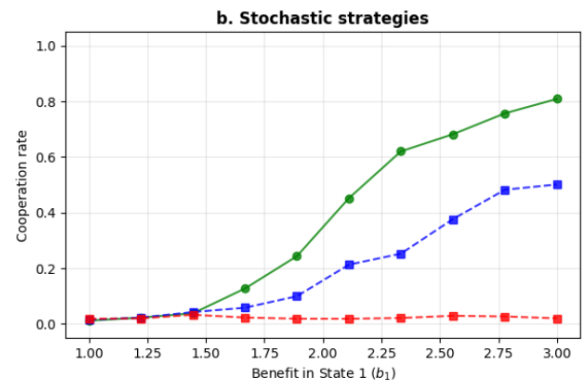

Figure 9: Reproduction of stochastic strategies with benefit in state 1

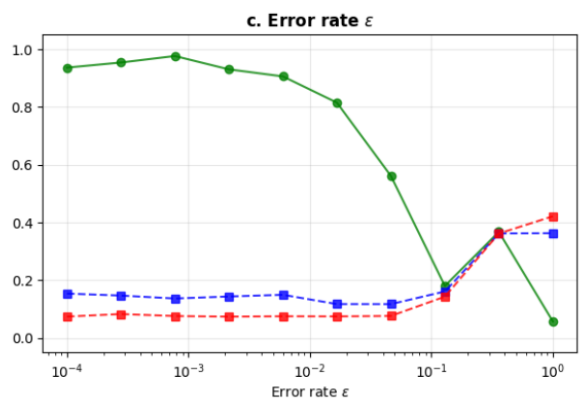

Figure 10: Reproduction with the error rate ε

of successful coordination deteriorates, causing the population to drift into the less valuable game (State 2). This demonstrates that the evolutionary advantage of stochastic games requires errors to be "sufficiently rare" for the feedback loop to remain stable. There is

a slight difference happens when there is a decrease when reaching 1, but in the paper, the authors didn't plot till there.

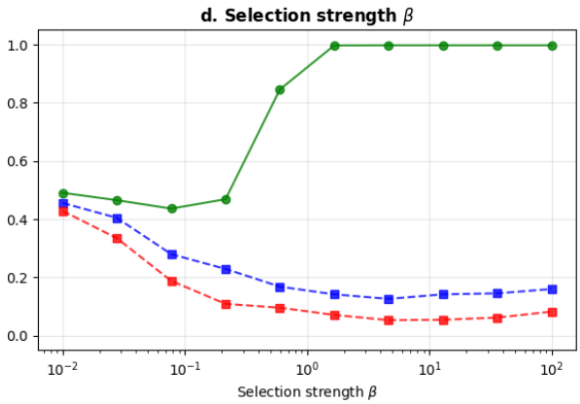

Figure 11: Reproduction with selection stength β

The reproduction shows similarities with the original plot, which demonstrates that evolution of cooperation in stochastic games is highly sensitive when varying the selection strength (*beta*).When it's low, the cooperation rate reaches 0.5, which shows that strategic differences have little impact. However, the more the selection strength increases, there is a sharp increase for the cooperation rate for the stochastic game compared to the fixed environment. This confirms what the paper proved, that when the environment's state depends on the players' actions, it greatly enhances the propensity for cooperation.

5.6 Analysis of Deterministic Transition Rules (Extended Data Fig. 2)

To understand which specific environmental feedback mechanisms successfully promote cooperation, we tested all eight deterministic transition structures. We map the next state to the current round's outcome: Mutual Cooperation, Mixed, or Mutual Defection. We conducted these simulations averaging the results over 100 independent runs for each rule (**a**–**h**).

We performed evolutionary simulations for each rule (labeled **a**–**h**). The results, averaged over 100 runs, are presented in Figure 12.

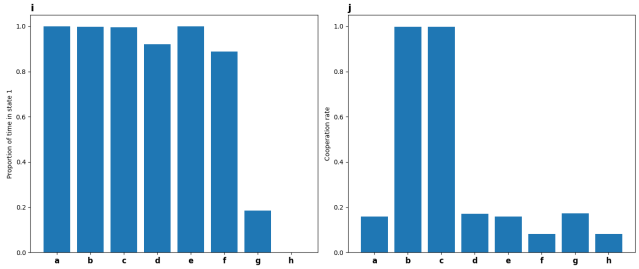

Figure 12: Impact of Transition Structure on Cooperation.

5.6.1 Results and Interpretation. One interesting thing we found is that a good environment doesn't guarantee cooperation. Even though rules a, d, e, and f keep the population in the 'Good State' nearly all the time, the actual cooperation rate is very low—less than 20 percent. Take Rule a for example: the environment stays good even if everyone cheats. Since the environment never crashes, there's no reason for players to stop being selfish. They just enjoy the high resources without helping out

So, what makes a rule successful? We found that only Rules b and c reach high cooperation levels. The reason is simple:Cooperation keeps the resources high, while mutual defection causes the environment to crash. This forces players to think about the future. If a rule is too 'forgiving,' like Rule a, people just take advantage of it. If it's too 'punishing,' like Rule h, players give up because cooperation doesn't save them anyway. To make cooperation evolve, the environment must act as an amplifier that rewards the group and punishes the selfish.

5.7 Evolving strategies analyze (Extended Figure 3)

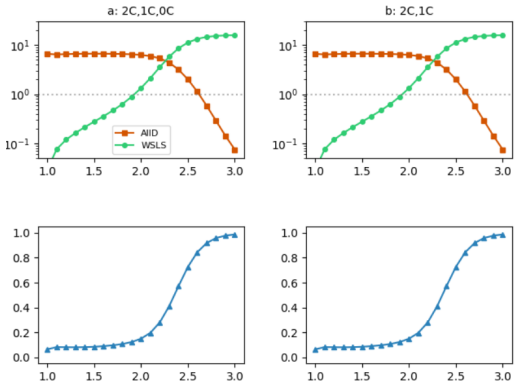

Figure 13: High Sensitivity to Feedback

In these cases, mutual cooperation significantly increases the probability of staying in or moving to the more beneficial State 1. The results show that the more the benefit increases, the more the cooperation rate climbs sharply. The upper plots reveal that WSLS quickly becomes the dominant strategy. This demonstrates that when players can earn a better environment through cooperation, WSLS helps to maintain that state.

Similar to the first two panels, these scenarios involve environments where cooperation leads to a good state. Even when the initial cooperation rate is low for small values of b1, there is a clear "tipping point" around b1 = 3. Once this threshold is crossed, WSLS becomes the most abundant strategy (indicated by the shaded bars). These panels reinforce the paper's core finding: the evolution of cooperation in stochastic games is largely driven by the success of WSLS-like strategies.

In these panels, the game parameters are set such that mutual cooperation does not effectively move the players toward the better state, or defection doesn't penalize them enough. Consequently, the lower plots show that cooperation rates remain stagnant or

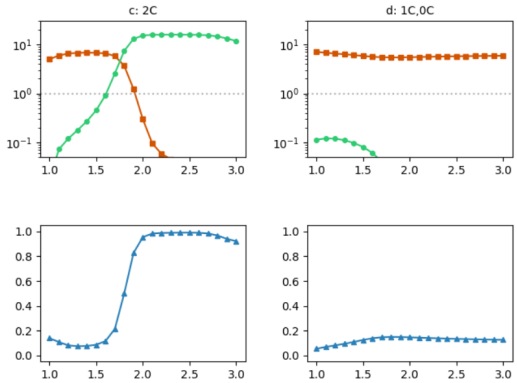

Figure 14: Feedback-Driven Cooperation

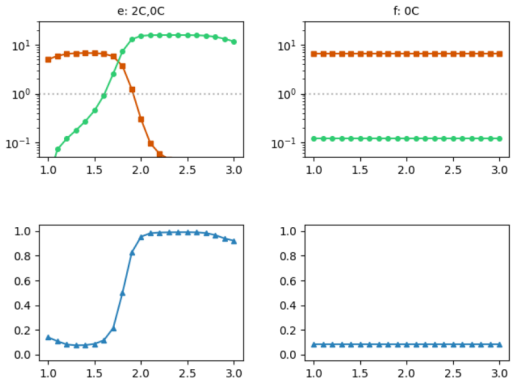

Figure 15: Decay of the Environment

low regardless of how high b1 becomes. The upper plots show that AllD remains the most abundant strategy throughout most of the parameter range, as there is no environmental incentive for players to switch to a reciprocal strategy.

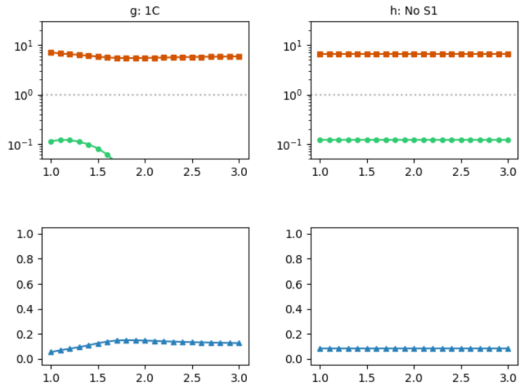

Figure 16: Resistance to Cooperation

These situations correspond to environments where the transition rules work against cooperation. For example, even when both players cooperate, there is still a high chance of ending up in the less rewarding State2. As a result, cooperation almost never appears. The strategy analysis shows that WSLS cannot establish itself, and the population stays dominated by AllD. This makes it clear that reciprocity on its own is not enough; the environment must also reinforce cooperation by making good states reliably follow cooperative behavior.

5.8 Impact of Transition Thresholds in Multiplayer Games (Extended figure 4)

We further investigated how the strictness of the environmental feedback rule influences cooperation. In our model, the environment transitions to (or remains in) the beneficial State 1 only if at least k players cooperate. If fewer than k players cooperate, the environment degrades to State 2.

To analyze this, we moved away from the stochastic simulations used in Figure 2 and instead utilized the analytical framework provided in run_final_simulation function. By calculating the exact fixation probabilities in the limit of rare mutations, we computed the stationary distribution for threshold values ranging from $k = 0$ to $k = 5$.

Figure 17 presents the results of this analytical calculation.

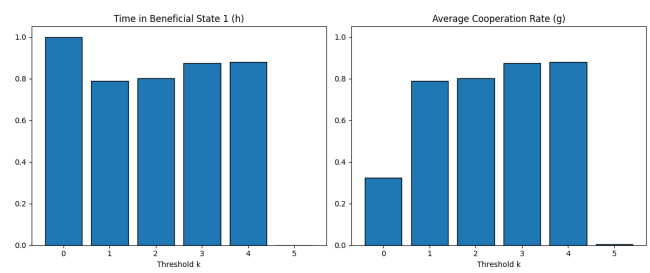

Figure 17: Reproduction of Extended Data Figure 4.

5.8.1 Analysis of Results. **The Free-Rider Problem ($k = 0$):** An interesting phenomenon occurs at $k = 0$. As shown in the left panel, the population spends 100% of its time in the beneficial State 1. However, the right panel reveals that the cooperation rate is low (≈ 0.3).

- *Explanation:* Since $k = 0$ implies the environment stays good regardless of player actions, there is no penalty for defection. Consequently, defectors invade the population, enjoying the high payoffs of State 1 without contributing. This confirms that a guaranteed good environment does not create cooperation.

Optimal Strictness ($k = 3, 4$): Cooperation peaks when the threshold is strict. For $k = 4$, the rule requires everyboidy to cooperate to maintain the public resource.

- *Explanation:* Our code implements this in calc_pay_numba via the condition next_env_val = 0 if nr_coop >= k. Under this strict rule, a single defector causes an immediate collapse to the low-payoff state. Because punishment is immediate, doing what is best for the group becomes what is best for the individual. This pushes the cooperation rate to nearly 90

System Collapse ($k = 5$): For $k = 5$, the threshold exceeds the group size ($N = 4$). This represents an impossible condition where the environment can never improve. As predicted, the system stays permanently in State 2 (Left Panel = 0), and cooperation collapses entirely (Right Panel ≈ 0).

6 Discussion

Our reproduction of the study validates the central hypothesis: cooperation thrives when the environment is responsive to player actions. By coupling game payoffs with environmental feedback, we identified three critical insights regarding the resulting evolutionary dynamics.

6.1 The Necessity of Environmental Consequences

The contrast between our stochastic and static simulations highlights that high benefits alone are insufficient to sustain cooperation. In the static control runs, despite a high benefit to cost ratio ($b = 2.0$), cooperation inevitably collapsed because there was no long-term penalty for defection. The success of the stochastic model proves that the future is an essential driver of collective action.

6.2 Discussion of probabilistic transitions maximize cooperation in three different stochastic games.

The reproduced figures illustrate how probabilistic transitions in stochastic games significantly influence the evolution of cooperation and average payoffs. While standard repeated games often fail to support cooperation, stochastic games introduce environmental feedback where improved conditions depend on player behavior. The results demonstrate that intermediate transition probabilities (q) often maximize payoffs. If the probability of restoring a good environment (Figure 3) or ending a bad interaction (Figure 4) is too low, cooperation fails to take hold; conversely, if restoration is too easy, defectors exploit the system. Consequently, a "sweet spot" of environmental uncertainty exists that incentivizes players to adopt cooperative strategies, such as "Win-Stay, Lose-Shift" (WSLS) or unconditional cooperation (All C).

6.3 Discussion: The Role of Environmental Strictness

Our analysis of the transition thresholds (k) offers a critical insight: a guaranteed good environment is actually bad for cooperation. Cooperation peaks only when the rules are strict. In this scenario, a single defector causes the environment to collapse. This fragility is actually a feature, not a bug: it forces individuals to cooperate to protect their own future payoffs. So, for cooperation to evolve, the environment must not only be beneficial but also fragile enough to punish non-compliance immediately.

7 Conclusions

Our simulations successfully reproduced the central findings of the original paper, confirming that environmental feedback is a critical driver for the evolution of cooperation.

- **The Power of Feedback:** Across both pairwise and public goods games (Fig. 2), linking game payoffs to environmental states increased cooperation rates to over 90%, whereas static environments consistently failed to sustain cooperation.
- **Responsiveness over Stability:** Our analysis of transition rules (Ext. Fig. 2) and thresholds (Ext. Fig. 4) revealed a key insight: a stable "good" environment does not guarantee cooperation. If the environment remains beneficial regardless of player actions (e.g., Rule **a** or threshold $k = 0$), free-riding dominates. Cooperation only evolves when the environment is *fragile*—meaning it degrades strictly and immediately in response to defection.
- **Strategic Dynamics:** We confirmed that immediate feedback (Fig. 4) and strong selection pressure are essential, especially in larger groups. Under these optimal conditions, the population converges on reciprocal strategies like "Win-Stay, Lose-Shift" (Ext. Fig. 3), establishing a robust loop between cooperative behavior and environmental quality.

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